

LGM: A Calibrated, Certifiable Factorized Neural Hawkes Process for Limit-Order-Book Simulation

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Abstract

Neural marked temporal point processes (MTPPs) predict the next limit-order-book (LOB) event well but *simulate* order flow poorly: in free roll-out they over-disperse or explode, and their stability cannot be honestly certified. We introduce **LGM** (Linear Ground $\times$ soft-max Marks), which factorizes the per-type intensity into a *linear* scalar ground rate and a *deep* soft-max mark head. Because the marks live on the probability simplex they are rate-neutral, so the total rate remains a pure linear Hawkes process regardless of mark-net nonlinearity. This preserves the exact stationary-mean formula, which we *pin* analytically to solve the train/simulate calibration mismatch, and yields a gauge-free branching ratio as a closed-form stability certificate. On Gemini ETH-USD data (62 event types), LGM is simultaneously calibrated (free-roll rate within $\sim 7\%$ of real), a competitive predictor, clustered (correct Fano profile), and certifiable — the first model in our study to achieve all of these together.

1 Introduction

Order-book simulation underpins market-making research, backtesting, and reinforcement learning of execution policies. [1]

2 The LGM factorization

The per-type intensity factorizes into a timing rate and a mark distribution,

$$\lambda_k(t) = \Lambda(t) p(k | t), \quad \sum_k \lambda_k(t) = \Lambda(t), \quad (1)$$

where the ground rate is a multi-timescale *linear* Hawkes process and the marks are a deep soft-max:

$$\Lambda(t) = \mu_0 + \sum_m a_m s_m(t), \quad p(\cdot | t) = \text{softmax}(z_\theta(t)). \quad (2)$$

Since the soft-max lies on the simplex, $\sum_k p_k = 1$, the marks are *rate-neutral*: the total rate Λ is unaffected by mark-net nonlinearity. The stationary mean therefore survives in closed form,

$$\bar{\Lambda} = \frac{\mu_0}{1 - n}, \quad n = \sum_m \frac{a_m}{\beta_m}, \quad (3)$$

and we **pin** it by setting $\mu_0 = R_{\text{target}}(1 - n)$, so $\bar{\Lambda} = R_{\text{target}}$ by construction. The branching ratio n is gauge-free and serves as a stability certificate, with $\text{Fano}(\infty) = 1/(1 - n)^2$.

Table 1: Headline comparison. LGM is uniquely calibrated *and* a competitive predictor *and* clustered *and* certifiable.

Model	GenAcc	Fano(1s)	branching ρ	free-roll rate	sim status
Compound Hawkes	0.18	8.8	0.64 (exact)	~ 2	stable, no long-memory
s2p2 (best predictor)	0.32	41	gauge-broken	—	over-disperses
NMH / GMH (windowed)	0.26	explode	1300 / 0.8	52–100/s	explode / Poisson
LGM ($n=0.86$)	0.29	6.0	0.86 (honest)	2.22/s	calibrated

3 Results

Table 1 reports the headline comparison on Gemini ETH–USD (62 event types, real rate ≈ 2.38 ev/s).

4 Discussion

References

- [1] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.